

Inconel 617 | High-Temperature Strength & Oxidation-Resistant Alloy

Introduction:

Inconel 617 (UNS N06617 / W.Nr. 2.4663) is a nickel-chromium-cobalt-molybdenum alloy designed for **exceptional high-temperature strength, oxidation, and carburization resistance**. This alloy maintains stability and mechanical integrity up to 1100°C, making it ideal for extreme service conditions in **gas turbines, chemical processing, and power generation systems**. It offers excellent weldability, formability, and corrosion resistance in both oxidizing and reducing environments.

Chemical Composition:

Element	Composition (%)
Nickel	44.5 min
Chromium	20.0 - 24.0
Cobalt	10.0 - 15.0
Molybdenum	8.0 - 10.0
Iron	3.0 max
Carbon	0.05 - 0.15
Manganese	1.0 max
Silicon	1.0 max
Sulfur	0.015 max
Titanium	0.6 max
Copper	0.5 max
Boron	0.006 max

Physical & Mechanical Properties:

- Density: 8.36 g/cm³
- Melting Range: 1332-1380 °C
- Tensile Strength: 734-769 MPa
- Yield Strength: 318-383 MPa
- Elongation: 56-62%
- Hardness: 170-190 BHN

Available Products:

- Seamless & Welded Pipes (ERW & EFW)
- Tubes (Capillary, U-Bend)
- Sheets, Plates, Coils & Strips
- Round, Flat, Hex & Square Bars
- Pipe Fittings, Flanges & Fasteners
- Custom Fabricated Components

Applications:

- **Aerospace & Gas Turbines:** combustion liners, transition ducts, and turbine components
- **Power Generation:** recuperators, heat exchangers, steam generator tubing
- **Petrochemical Plants:** reformers, flare tips, catalyst grids, furnace internals
- **Industrial Furnaces:** radiant tubes, trays, retorts, heat shields
- **Waste Incineration & Pollution Control:** liners, baffles, exhaust ducts
- **Chemical Processing:** reactors, piping, heat treatment components
- **Heat Recovery Systems:** recuperators and high-efficiency heat exchangers

Advantages:

- Excellent oxidation and carburization resistance up to 1100°C
- High tensile and creep strength at elevated temperatures
- Outstanding resistance to nitriding and corrosive gases
- Excellent metallurgical stability and weldability
- Long service life under thermal cycling and high stress

Why Choose Moksh Tubes & Fittings LLP:

- Specialists in nickel and high-temperature alloys
- Global supply with ASME-approved materials
- Custom fabrication with strict quality control
- Reliable delivery and technical support